Bentley iTwin Capture Modeler

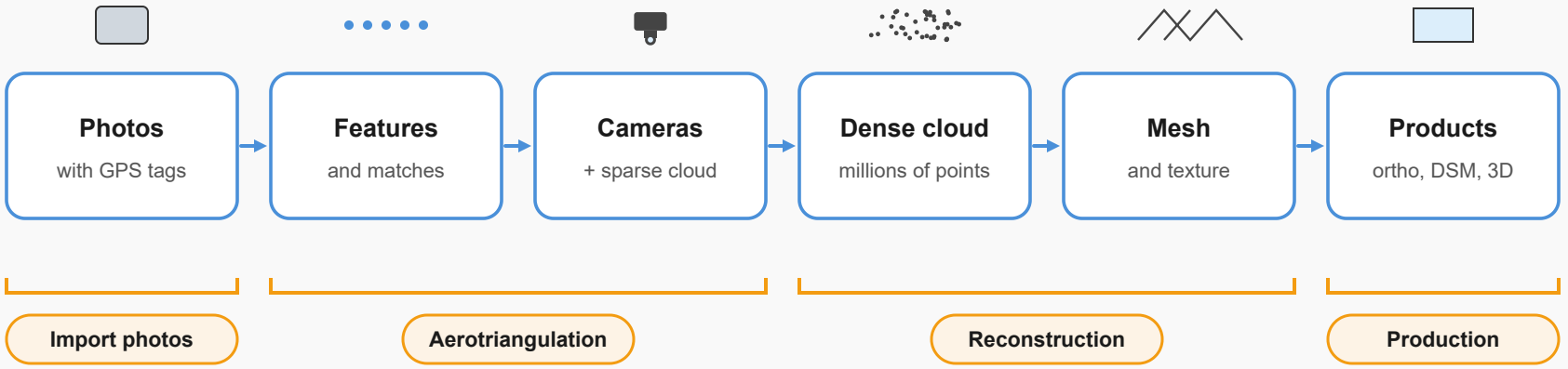
Processing Lab — step by step

Today

- Photos in, orthophoto and 3D model out
- Steps: project, import, aerotriangulation, reconstruction, production
- Then open the orthophoto in QGIS
- What to check along the way
- What to turn in

Six stages, four buttons

What structure from motion does, and what the software calls each part of it.



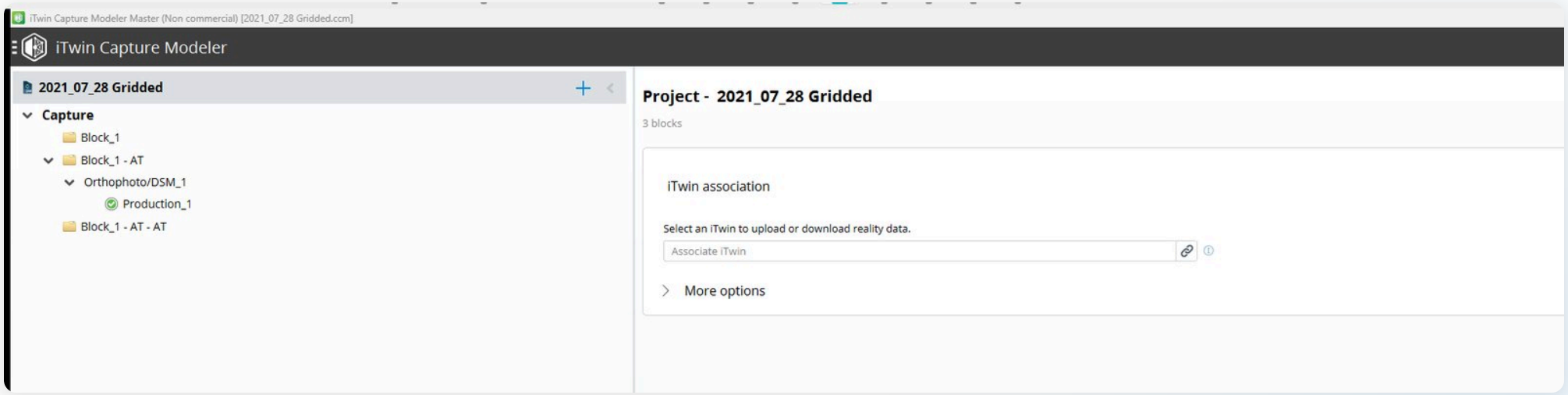
Bentley iTwin Capture Modeler, the four steps you click

The software does the arithmetic. The photos decide whether the arithmetic has anything to work with.

Six stages, four buttons. Import, Aerotriangulation, Reconstruction and Production map onto the pipeline you saw Tuesday.

Step 1 — Create a Project

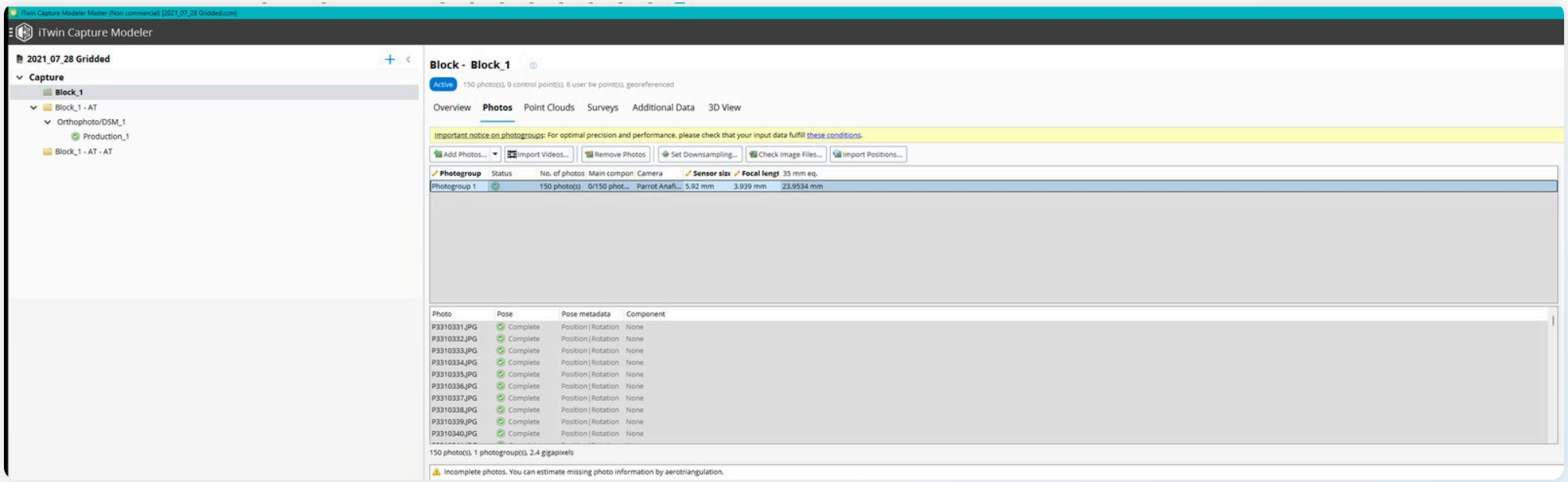
- Open **iTwin Capture Modeler**
- Create a **new project**; name it and choose the project folder
- **New Capture**, then **New Block**: the set of photos
- Engine must be running, or jobs stay **pending**



One project: a block of photos, its aerotriangulated copy, a reconstruction and a production. iTwin Capture Modeler, Bentley Systems.

Step 2 — Import Photos

- Open the block's **Photos** tab
- Add all the photos of a directory
- GPS positions are read from the photo metadata
- Check the block **3D view**: do the cameras match your flight?



The block's Photos tab, with the buttons for adding photos. iTwin Capture Modeler, Bentley Systems.

The Bad Photo Hunt

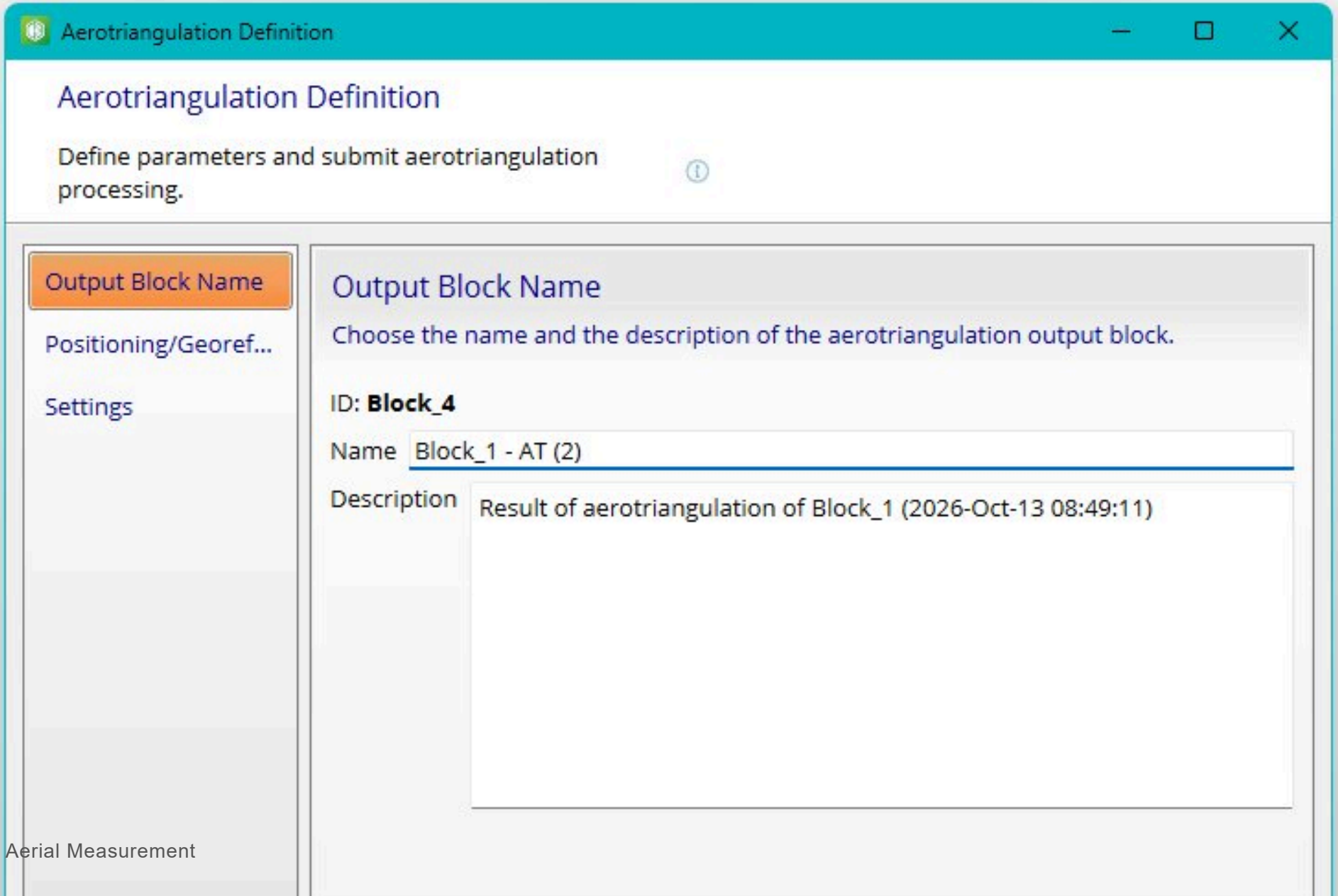
Before the next step, scan your photos. **Delete** any that are:

- Blurry
- Pointed at the ground during takeoff
- Far from the study area

Garbage in, garbage out.

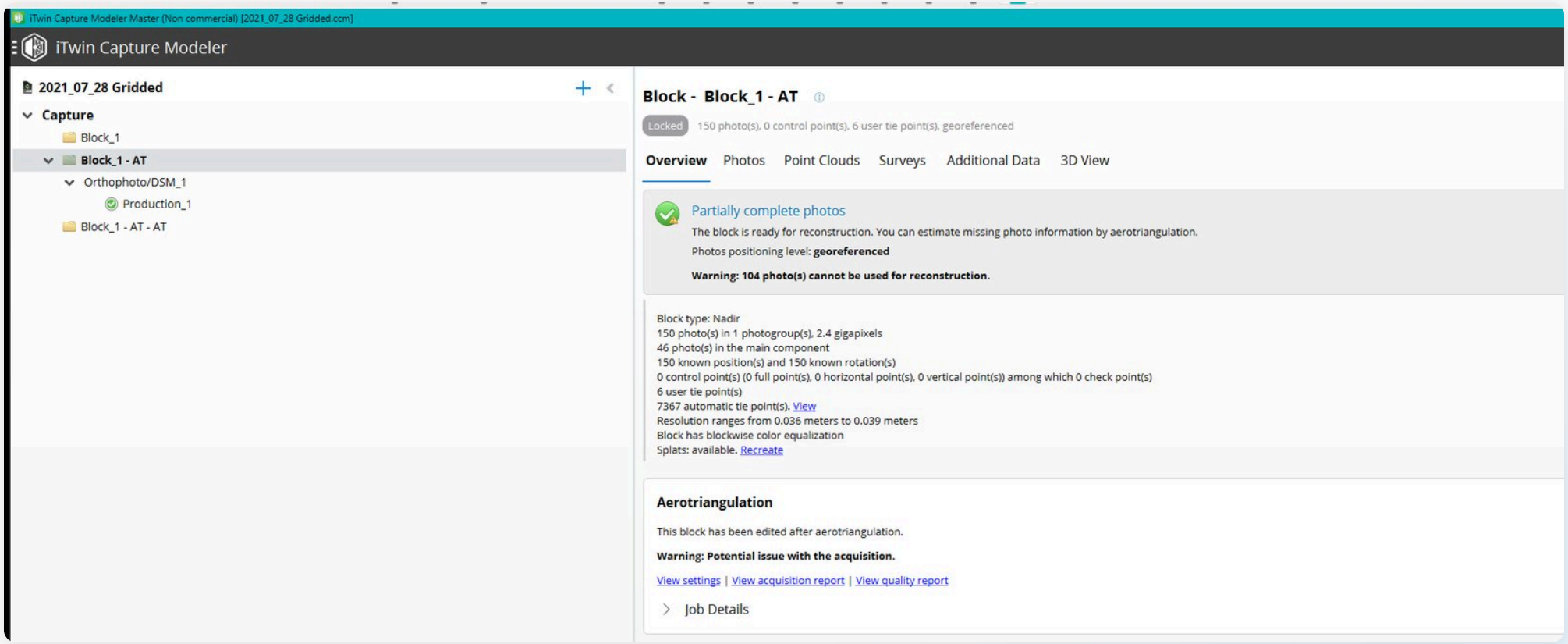
Step 3 — Aerotriangulation

- Click **Submit Aerotriangulation**, then choose the **Engine**
- Keep the **default settings** (Adjust, Normal, Default)
- Click **Submit**, then wait
- Result: a new block with a **sparse point cloud**

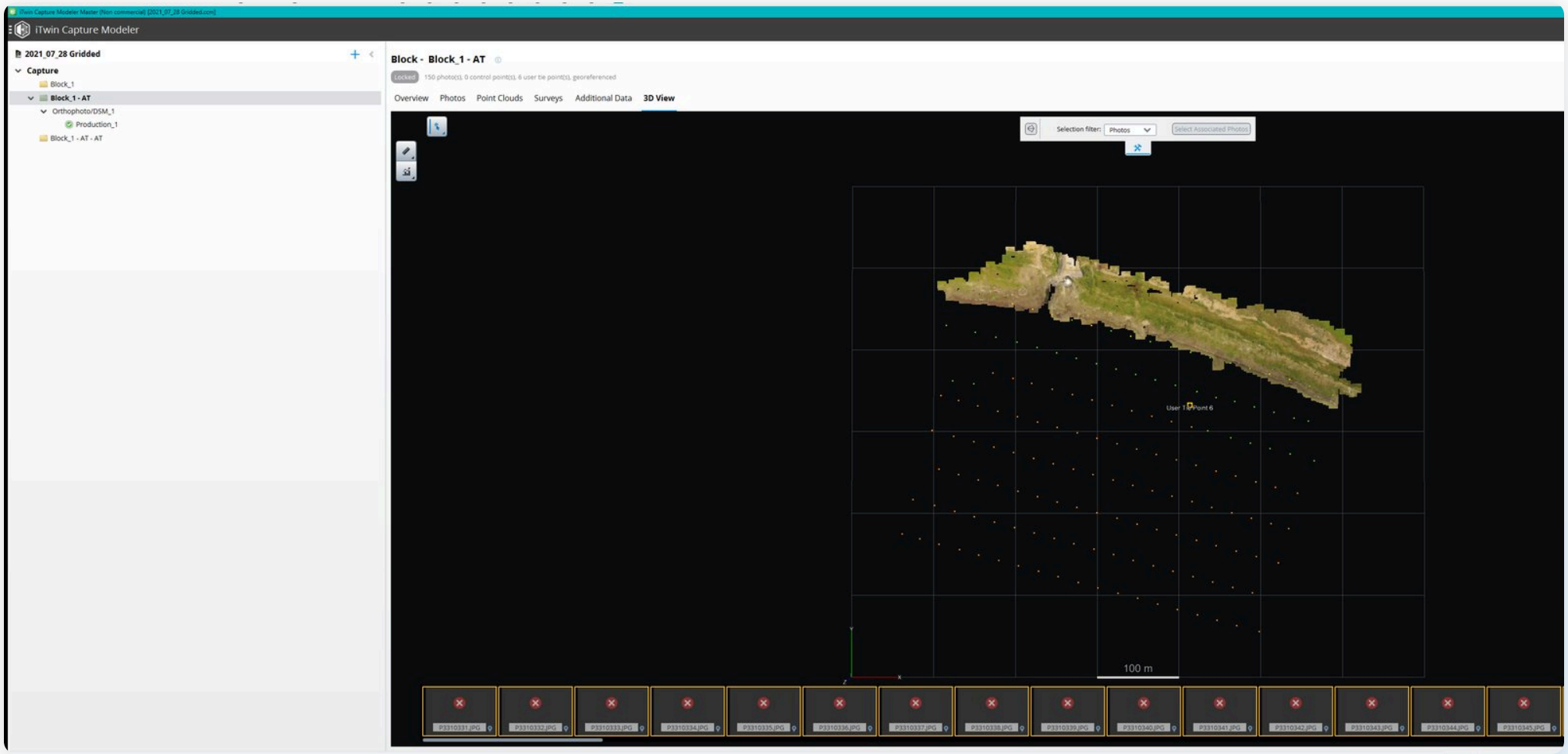

Aerotriangulation Definition
Aerotriangulation Definition
Define parameters and submit aerotriangulation processing.
Output Block Name
Positioning/Georef...
Settings
Output Block Name
Choose the name and the description of the aerotriangulation output block.
ID: Block_4
Name Block_1 - AT (2)
Description
Result of aerotriangulation of Block_1 (2026-Oct-13 08:49:11)

Reading the AT result

- **Photos calibrated:** all, or nearly all
- **View quality report:** reprojection error under 1 px is good
- Sparse cloud, **3D view**: most points on brick and grass
- **Holes:** shiny surfaces, windows, fresh asphalt



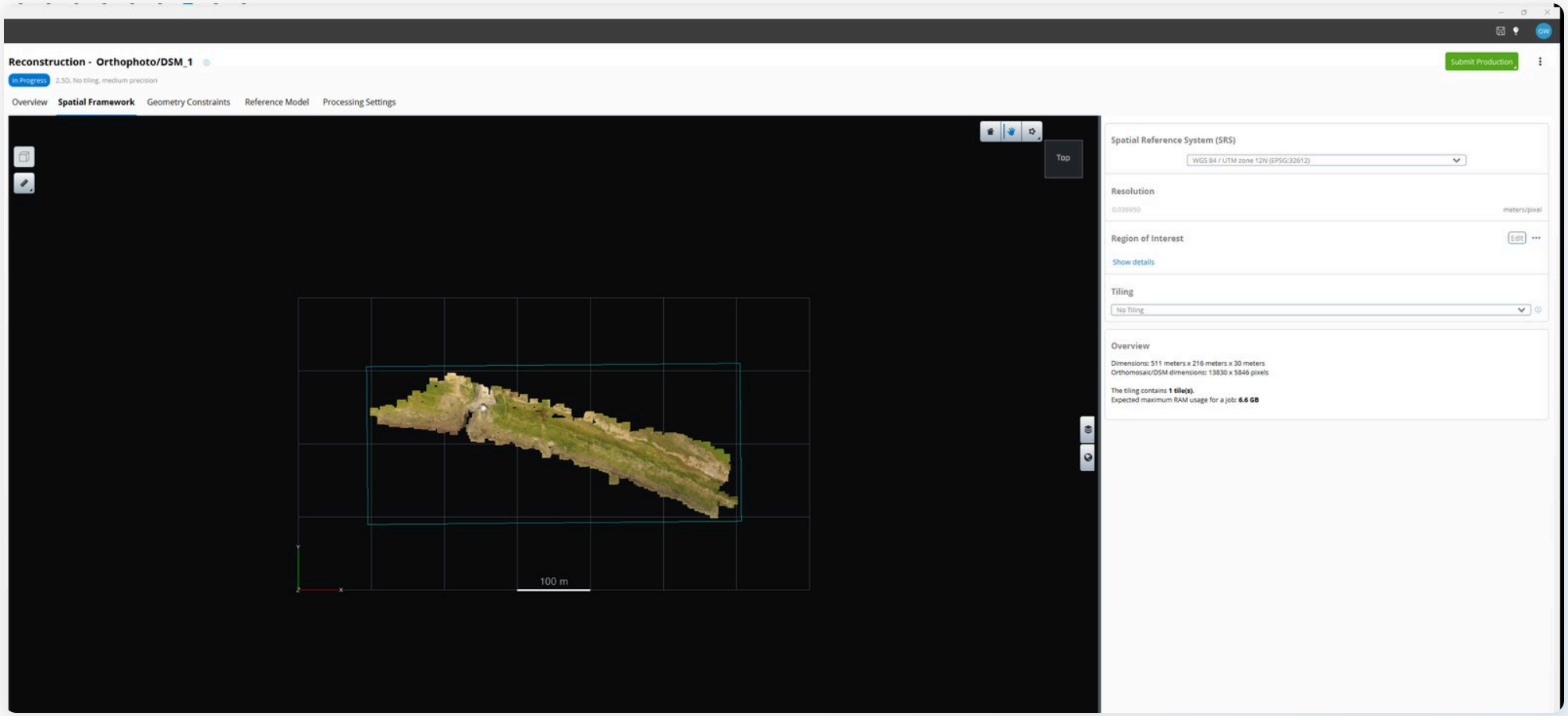
The output block's dashboard: how many photos were calibrated. iTwin Capture Modeler, Bentley Systems.



The sparse cloud and the camera positions in the output block's 3D View. iTwin Capture Modeler, Bentley Systems.

Step 5 — Reconstruction

- (Step 4, ground control, is skipped in this lab)
- **New Reconstruction**: **Orthophoto/DSM** for the map, **3D reconstruction** for mesh and points
- **Spatial Framework:** set `EPSG:32612`, check the region of interest
- **Geometric precision:** Medium for the map



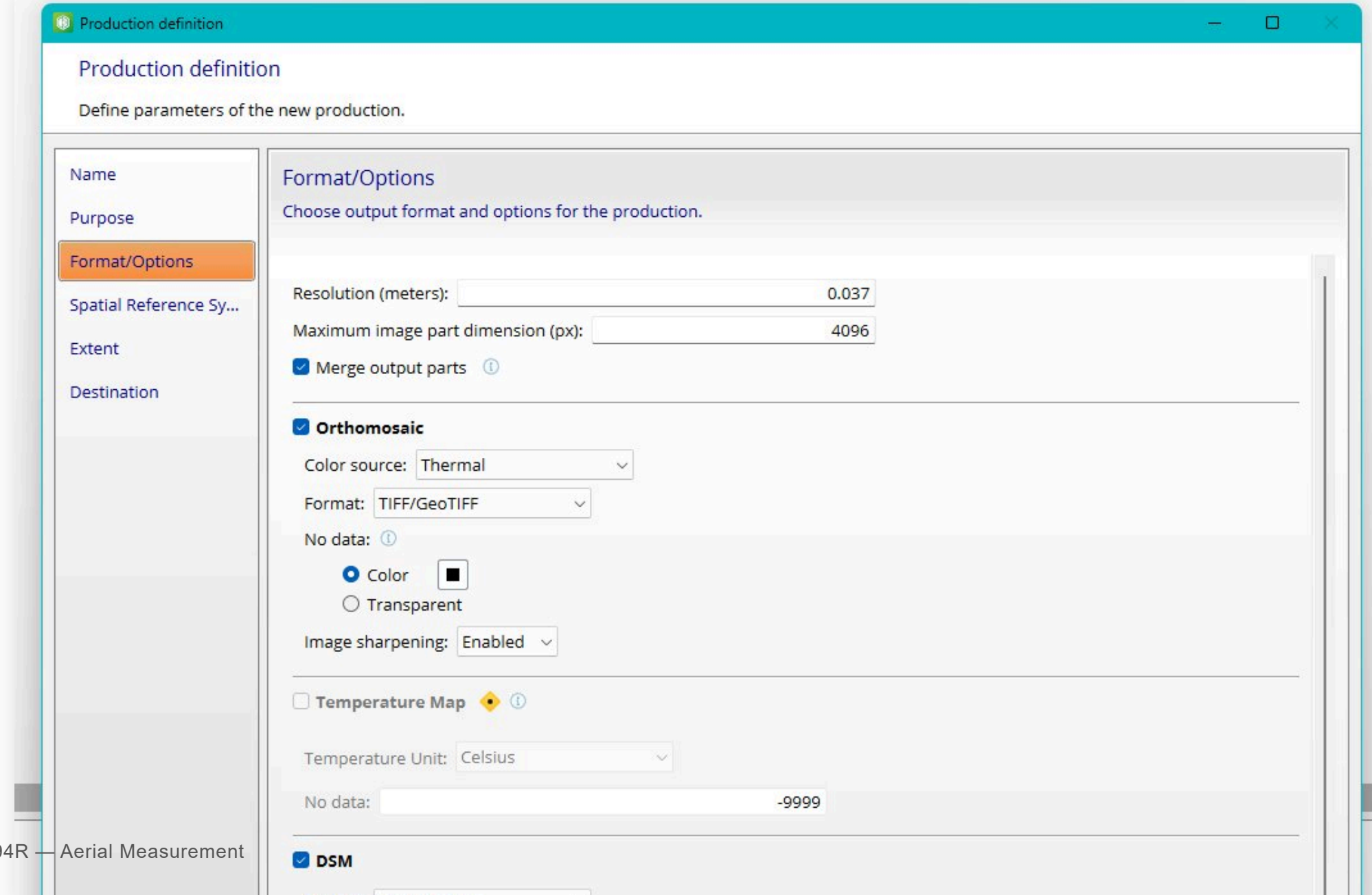
The Spatial Framework tab: reference system, resolution and region of interest. iTwin Capture Modeler, Bentley Systems.

Step 6 — Production

Click **Submit Production**, pick the purpose:

- **Export Orthomosaic/DSM** — **TIFF/GeoTIFF**, `EPSG:32612`, **Resolution** in meters
- **Export 3D mesh** — **iTwin Ready** or **OBJ wavefront**
- **Export 3D point cloud** — **LAS**

QGIS needs the orthophoto GeoTIFF.


Production definition
Production definition
Define parameters of the new production.
Name
Purpose
Format/Options
Spatial Reference Sy...
Extent
Destination
Format/Options
Choose output format and options for the production.
Resolution (meters): 0.037
Maximum image part dimension (px): 4096
Merge output parts
Orthomosaic
Color source: Thermal
Format: TIFF/GeoTIFF
No data:
Color
Transparent
Image sharpening: Enabled
Temperature Map
Temperature Unit: Celsius
No data: -9999
DSM

While it runs

- **Aerotriangulation is the brain.** If it fails, the model fails
- **Sparse** cloud comes first: points only
- **Dense** reconstruction, the actual model, comes after
- Look at the sparse cloud now: where are the holes?

The sparse cloud has a hole over the parking lot.

What happened on the flight?

What to turn in

The lab page has the deliverables: Data Processing.

Full written steps: Bentley iTwin.